

SIMCITY: THE WORST GAME RELEASE IN HISTORY

A game release hogging the headlines for all the wrong reasons. It's every publisher's worst nightmare. Everything that possibly could go wrong went wrong in the much-awaited SimCity title by EA. A roundup of what transpired.
<http://dgit.in/15VkYIM>

OPINION: IS SCIENTIFIC TRUTH ALWAYS BEAUTIFUL?

According to a mathematician, too many researchers pursue the quest for symmetry in scientific discovery. Science has a serious "beauty" problem and may not always be elegant.
<http://dgit.in/YBQINI>

THE GIANT SQUID

The stuff of legends and persistent human

fascination, the giant squid's been finally caught on film for the very first time in its natural environment. Watch what it took to lure the silent sea monster to the camera.
<http://dgit.in/1073vo0>

THE INSIDE STORY OF THINKPAD'S REDESIGN

Revamping the most iconic (and loved) laptop lineup in history takes one heck of a balancing act. Change too much and your fans crib, change too little and your rivals maintain the edge. So how exactly did Lenovo redesign the ThinkPad?
<http://dgit.in/134lvSd>

After 25 years, NASA restarts plutonium production

- By Joel Hruska



Copyright © 2012
Ziff Davis Publishing
Holdings Inc.

After a quarter-century hiatus, the United States has begun producing plutonium-238 once more. The decision was made to ensure that future NASA projects would have access to the valuable fuel. As US stocks dwindled, NASA began buying plutonium-238 from Russia, but that agreement came to an end in 2010. When most people think of plutonium, they think of nuclear weapons — but that's not what plutonium-238 is used for.

If you need a power source that can last for decades, plutonium-238 is fantastically useful stuff. It's got a half life of nearly 88 years and it emits 560 watts of heat per kilogram of material. It's a vital component of the radioisotope thermoelectric generators (RTGs) used on Curiosity and in a number of space probes, including Cassini.

One of the best features of plutonium-238 is that, while it's radioactive as hell (275 times more so than plutonium-239, it takes a minimal amount of shielding to protect spacecraft or humans from contamination. Plutonium-powered pacemakers (yes, that was a thing for a little while) have operated as long as 25 years without running out of power.

The schedule drawn up by NASA and the Department of Energy doesn't call for much production; the organization is hoping to create approximately 1.5 kilograms (3.3 lbs) a year. NASA has found



that it can hit its energy density targets by mixing some of its older plutonium stock at a ratio of 1:2 — another important factor. One of the most basic challenges of building probes and spacecraft for deep space exploration is the need to keep the vehicles supplied with enough power. In many cases, solar panels simply aren't an option — they're difficult to keep clean and their power output falls as a probe's distance from the Sun increases.

One of the upcoming missions that will require a plutonium-powered RTG is the Curiosity follow-up, which will see another Curiosity-like rover deposited on Mars, hopefully by 2020.

Increasing plutonium-238 production is only one of NASA's goals for Curiosity's follow-up. One of the downsides of RTGs is that the thermocouples they rely on to turn heat into electrical energy are extremely inefficient — on the order of 5-7%. This is why Curiosity, which carries an RTG that generates some 2000W of heat energy, only has a meager 125W of electrical power

to work with. RTGs also degrade slowly over time — Voyager's power levels have dropped to 67% due to a combination of radioactive decay and thermocouple aging. If not for the latter, available power would still be in the 80% range.

NASA hopes to significantly expand the power budget of future missions by including an Advanced Stirling Radioisotope Generator, or ASRG. This type of generator has been in testing for years, and can hit efficiencies of 20% or more. That would more than triple the available power without requiring a dramatic increase in probe weight, radioactive shielding, or additional plutonium-238.

According to NASA officials, the change isn't entirely positive — Curiosity recycles a great deal of waste heat to keep the rover's electronics at a safe operating temperature, and the use of an ASRG would significantly reduce the amount of heat that could be pumped to other parts of the rover. Still, this is considered a solvable problem.

The upside benefits, meanwhile, would be enormous. More power means more options for processing power, higher bandwidth transmitters, and (we hope) a bigger laser. It would give mission planners more flexibility and open up longer mission windows. There's no plan to use the plutonium-238 in other ways — at least not one that's publicized. This initiative is separate from the 500MW molten salt reactors we covered recently, and isn't a part of the ongoing discussion of whether or not nuclear power should be on the table of environmental alternatives to traditional fossil fuels.